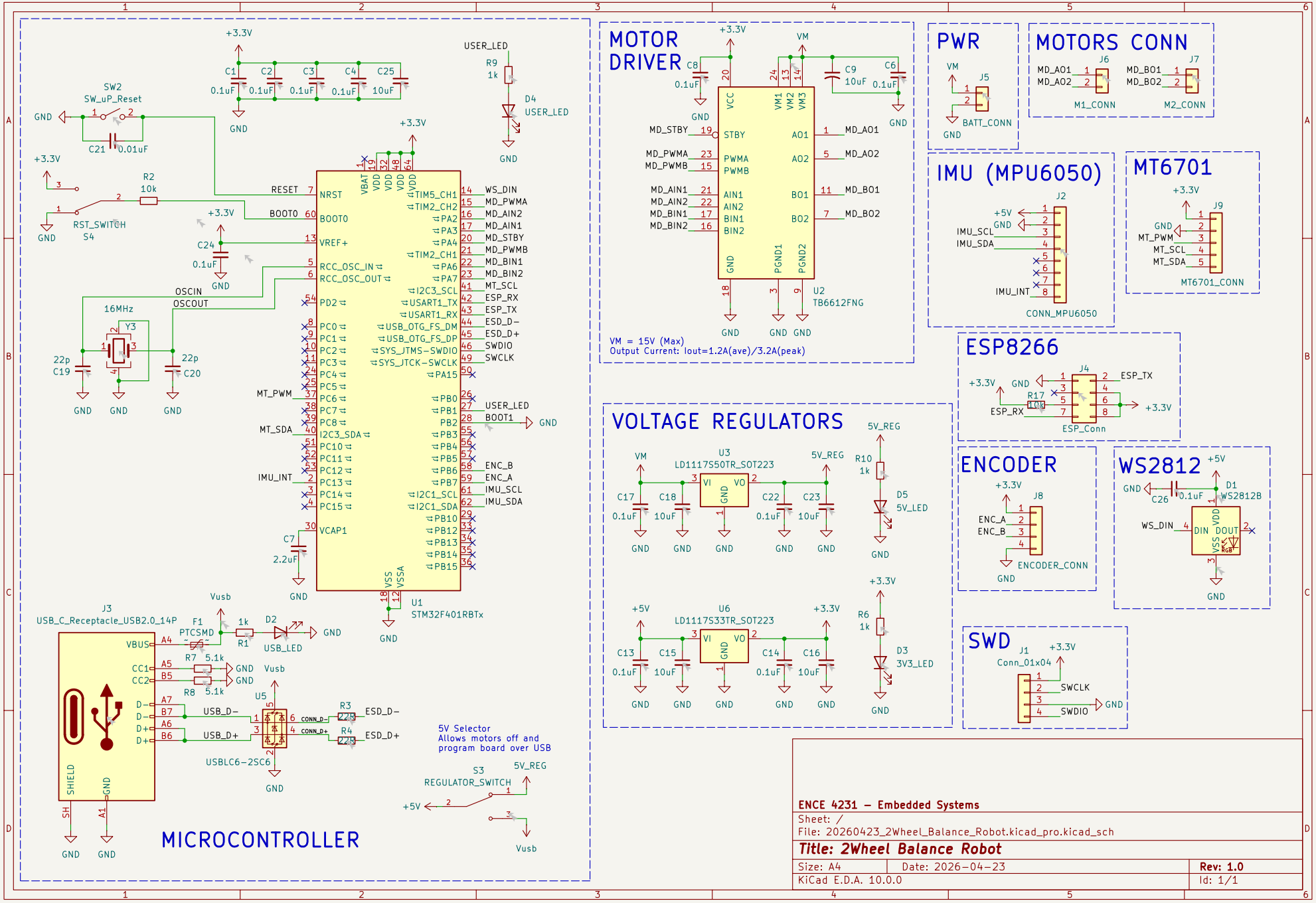
MICROCONTROLLER
MOTOR DRIVER
PWR
MOTORS CONN
IMU (MPU6050)
MT6701
ESP8266
ENCODER
WS2812
VOLTAGE REGULATORS
SWD
U1 STM32F401RBTx
U2 TB6612FNG
U3 LD1117S50TR_SOT223
U6 LD1117S33TR_SOT223
J3 USB_C_Receptacle_USB2.0_14P
U5 USBLC6-2SC6
S3 REGULATOR_SWITCH
SW2 SW_uP_Reset
RST_SWITCH S4
Y3 16MHz
J2 CONN_MPU6050
J4 ESP_Conn
J8 ENCODER_CONN
J9 MT6701_CONN
M1_CONN
M2_CONN
BATT_CONN
J1 Conn_01x04
D1 WS2812B
D2 USB_LED
D3 3V3_LED
D4 USER_LED
D5 5V_LED
VM = 15V (Max)
Output Current: Iout=1.2A(ave)/3.2A(peak)
5V Selector
Allows motors off and
program board over USB
ENCE 4231 – Embedded Systems
Sheet: /
File: 20260423_2Wheel_Balance_Robot.kicad_pro.kicad_sch
Title: 2Wheel Balance Robot
Size: A4
Date: 2026-04-23
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 1/1